EXPT:-07. FLOW CONTROL AT DATA LINK LAYER
13-08-2020. USING SLIDING WINDOW PROTOCOL.

AIM:-
To write a program to implement flow control at the data link layer using the sliding window protocol and to simulate the flow of frames from one node to the another.

ALGORITHM:.

1. Start the program.
2. Read the total number of frames to be sent(N), the sender's window size(W) and a frame no. to simulate as lost during transmission.
3. Initialize base=0 and nextSeq:=0.
4. Repeat while base<N:
   a). Determine the current sender window as the range [base, base+W-1].
   b) Send frames from nextSeq up to the window limit (base+W-1), incrementing nextSeq per frame.
   c) If a frame is lost: Reciever discards out-of order frames, sender times out and retransmits all frames from the lost frame onward (nextSeq = lost frame).
   d) Else (no loss): Reciever sends out cumulative ACKs; sender slides window forward.

```
=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (Go-Back-N) ===

Enter total number of frames to send      : 6
Enter sender's window size                : 3
Enter frame number to simulate as LOST
(0 to 5, enter -1 for no loss) : 3

-----------------------------------------
Sender Window : [Frame 0 - Frame 2]
Node A --> Sending Frame 0
Node A --> Sending Frame 1
Node A --> Sending Frame 2
Node B --> Received Frame 0, sending ACK 0
Node B --> Received Frame 1, sending ACK 1
Node B --> Received Frame 2, sending ACK 2
Node A --> ACK 2 received. Sliding window forward.

-----------------------------------------
Sender Window : [Frame 3 - Frame 5]
Node A --> Sending Frame 3
           (Frame 3 LOST in transit!)
Node A --> Sending Frame 4
Node A --> Sending Frame 5
Node B --> Frame 3 missing. Discarding out-of-order frames after it.
Node A --> Timeout! Resending from Frame 3.

-----------------------------------------
Sender Window : [Frame 3 - Frame 5]
Node A --> Sending Frame 3
Node A --> Sending Frame 4
Node A --> Sending Frame 5
Node B --> Received Frame 3, sending ACK 3
Node B --> Received Frame 4, sending ACK 4
Node B --> Received Frame 5, sending ACK 5
Node A --> ACK 5 received. Sliding window forward.

-----------------------------------------
All 6 frames successfully transmitted from Node A to Node B.
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5. Repeat step4 until all N frames have been transmitted and acknowledged.

6. Stop.

PROGRAM:-

```
def simulate(total_frames, window_size, lost_frame):
    base = 0
    next_seq = 0
    while base < total_frames:
        window_end = min(base+window_size-1,
            total_frames - 1)

        print("\n----------")
        print(f"Sender Window: [Frame{base}-Frame
            {window_end}]").

    while next_seq < base+window_size and
        next_seq < total_frames:
        print(f"Node A --> Sending Frame{next_seq
            )

        if next_seq == lost_frame:
            print(f" (Frame{next_seq} LOST in
                transit!)").

            next_seq += 1.

If lost_frame is not None and base <= lost_frame
    < next_seq:
    print(f"Node B -> Frame{lost_frame} missing.
    Discarding out-of-order frames after it").

    next_seq = lost_frame.
    lost_frame = None.
```

```
=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (Go-Back-N) ===

Enter total number of frames to send      : 8
Enter sender's window size                : 4
Enter frame number to simulate as LOST
(0 to 7, enter -1 for no loss) : 5

-----------------------------------------
Sender Window : [Frame 0 - Frame 3]
Node A --> Sending Frame 0
Node A --> Sending Frame 1
Node A --> Sending Frame 2
Node A --> Sending Frame 3
Node B --> Received Frame 0, sending ACK 0
Node B --> Received Frame 1, sending ACK 1
Node B --> Received Frame 2, sending ACK 2
Node B --> Received Frame 3, sending ACK 3
Node A --> ACK 3 received. Sliding window forward.

-----------------------------------------
Sender Window : [Frame 4 - Frame 7]
Node A --> Sending Frame 4
Node A --> Sending Frame 5
           (Frame 5 LOST in transit!)
Node A --> Sending Frame 6
Node A --> Sending Frame 7
Node B --> Frame 5 missing. Discarding out-of-order frames after it.
Node A --> Timeout! Resending from Frame 5.

-----------------------------------------
Sender Window : [Frame 5 - Frame 7]
Node A --> Sending Frame 5
Node A --> Sending Frame 6
Node A --> Sending Frame 7
Node B --> Received Frame 5, sending ACK 5
Node B --> Received Frame 6, sending ACK 6
Node B --> Received Frame 7, sending ACK 7
Node A --> ACK 7 received. Sliding window forward.

-----------------------------------------
All 8 frames successfully transmitted from Node A to Node B.
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```
retransmission succeeds.
        continue.
for f in range(base, next_seq):
    print(f"Node B --> Recieved Frame {f}, sending
                ACK {f}").
    print(f"Node A --> ACK {next_seq-1} recieved.
        Sliding window forwaid.").
        base = next_seq.
print("\n-----------").
print(f"All {total_frames} frames succesfully
        transmitted from Node A to Node B").

def main():
    print("=== SLIDING WINDOW PROTOCOL-FLOW
        CONTROL SIMULATION (Go-Back-N) ===\n").
total_frames = int(input("Enta total numbea of
        frames to send:")).
lost_frame = int(input(f"Enta frame numbea to
        simulate as LOST \n(0 to {total_frames-1}
        enter -1 for no loss):")).
if lost_frame == -1:
        lost_frame = None.
simulate(total_frames, window_size, lost_frame).
if __name__ == "__main__":
    main().
```

RESULT:-

The data link layer sliding window protocol was successfully implemented in python and verified through sender-reciever mode simulation.